



SPRING END SEMESTER EXAMINATION-2025
2nd Semester M.Tech/Ph.D Coursework

APPLICATION ARCHITECTURE & DEPLOYMENT
CS60508

(For 2024 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes

Full Marks: 50

Answer any FIVE questions.

Question paper consists of two SECTIONS i.e. A and B.

Section A is compulsory.

Attempt any Four question from Sections B.

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

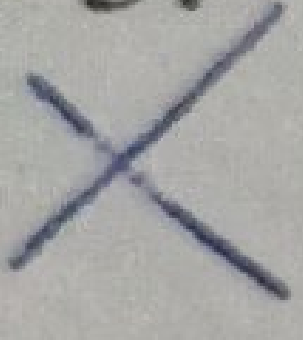
1. Answer the following questions.

[1 × 10]

- (a) How does microservices architecture differ from monolithic architecture?
- (b) What does API stand for in software development?
- (c) Which HTTP method is typically used to retrieve data from a server?
- (d) What is the difference between a Docker image and a Docker container?
- (e) What is the role of a Dockerfile in containerization?
- (f) Give a Dockerfile example
- (g) Why are YAML files used in Kubernetes configurations? Give one example
- (h) Define a 'Pod' in Kubernetes. How to manage pods in Kubernetes.

- (i) What does MLOps stand for in machine learning?
- (j) Name one stage of the Software Development Life Cycle (SDLC).

SECTION-B

- 2. (a) Compare and contrast monolithic and microservices architectures in terms of deployment, scalability, and fault isolation. [5]
- (b) Provide a real-world example of a system that transitioned from monolithic to microservices architecture. What challenges were encountered during this transition? [5]
- 3.  (a) Describe the purpose and typical structure of HTTP methods (GET, POST, PUT, DELETE) in RESTful APIs. [5]
- (b) What tools are commonly used for API testing, and what types of testing can they perform? [5]
- 4. (a) What is the difference between virtualization and containerization, and what are the advantages of using containers? [5]
- (b) Outline the lifecycle of a container from creation to termination. [5]
- 5. (a) What is Kubernetes, and why is it used in container orchestration? [5]
- (b) How does Kubernetes handle networking between containers and services within a cluster? [5]

6. (a) What is MLOps, and how does it integrate with the traditional Software Development Life Cycle (SDLC)? [5]
- (b) How does MLOps address challenges such as model versioning, monitoring, and continuous integration/continuous deployment (CI/CD)? [5]
